

SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering
Program: B.Tech/MBA TechAI Semester: VI

Course: Advanced topics in Machine Learning
Experiment No.07

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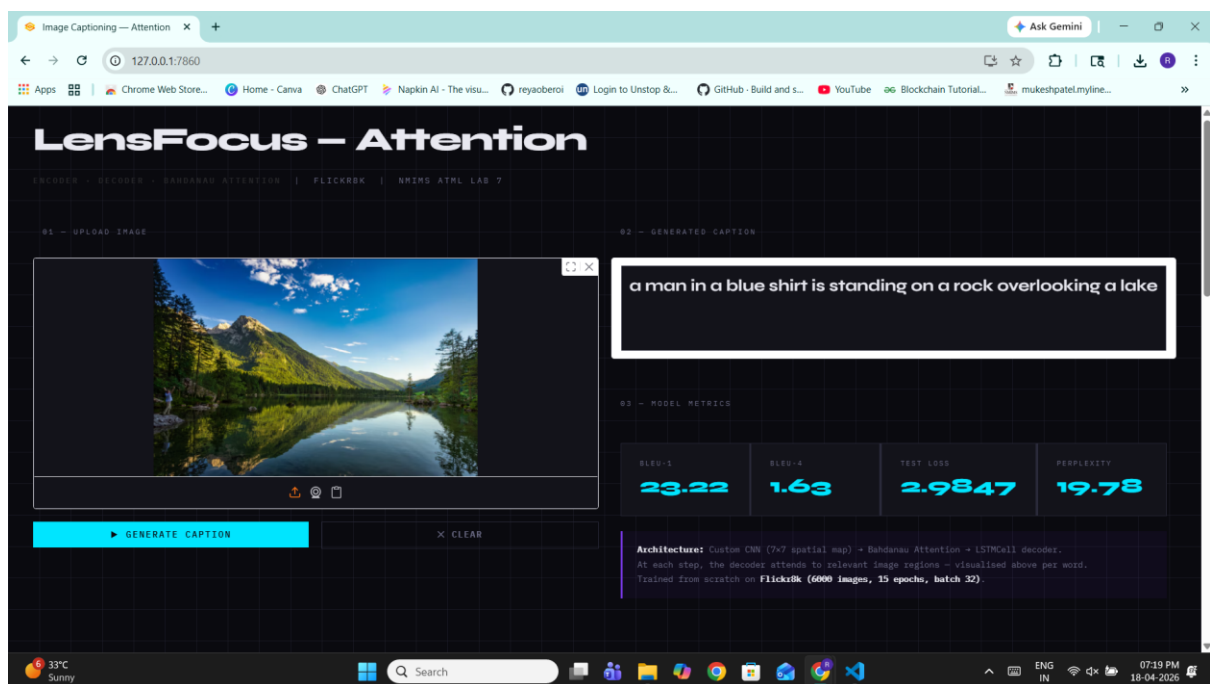
Image Captioning using Attention

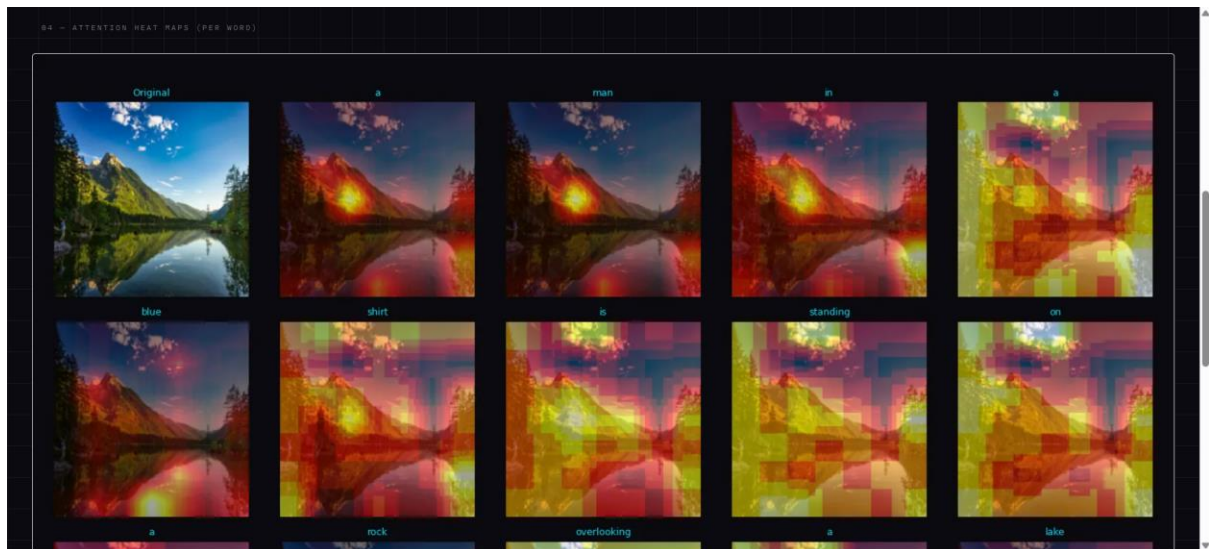
Encoder-Decoder with Bahdanau Attention on Flickr8k Dataset

Implemented an image captioning model using an Encoder–Decoder architecture with Bahdanau Attention on the Flickr8k dataset. The model generates natural language descriptions for images and visualizes attention maps for interpretability. The trained model is deployed using a Gradio-based frontend interface.

Github repository: <https://github.com/reyaoberoi/image-captioning-attention.git>

Final Output (Frontend Result)



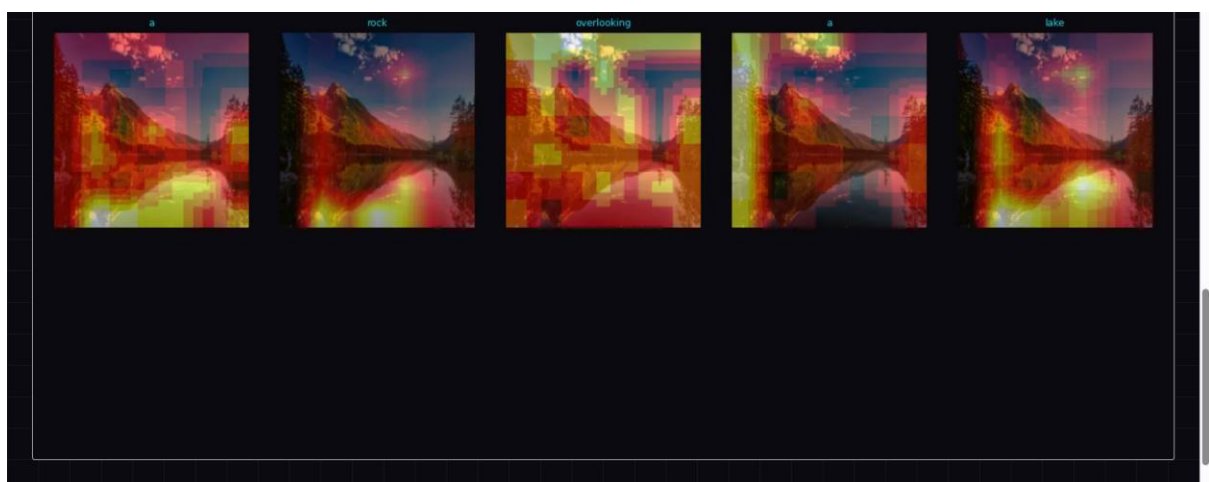


The interface allows users to upload an image and generate a descriptive caption using a trained attention-based deep learning model. It also displays evaluation metrics such as BLEU score and perplexity, along with attention heatmaps that visualize how the model focuses on different parts of the image while generating each word.

The model performance is evaluated using four key metrics displayed in the interface. Test Loss and Perplexity measure how well the model predicts the next word, while BLEU-1 and BLEU-4 evaluate the quality of generated captions by comparing them with ground truth captions.

Compared to the encoder-decoder model without attention, the attention-based model shows improved BLEU scores and better contextual understanding. The attention mechanism helps the model focus on relevant image regions, leading to more accurate and meaningful caption generation.

Lower perplexity and improved BLEU scores indicate that the attention model generates more fluent and contextually relevant captions.



Notebook Explanation & Training Process

- Loaded Flickr8k dataset and preprocessed images and captions
- Tokenized captions and created vocabulary mappings
- Built CNN-based encoder to extract spatial image features (7×7 feature map)
- Implemented Bahdanau Attention to focus on relevant image regions
- Designed LSTM decoder to generate captions word-by-word
- Used embedding layer for converting words to vectors
- Applied teacher forcing during training for better convergence
- Loss function: Cross-Entropy Loss
- Optimizer: Adam

Training Details

Dataset size: ~6000 images

Epochs: 15

Batch size: 32

4. Hyperparameters

```
DATASET_SIZE    = 6000
BATCH_SIZE      = 32
NUM_EPOCHS      = 15
LEARNING_RATE   = 3e-4
EMBED_DIM       = 256
HIDDEN_DIM      = 512
NUM_LAYERS      = 1
IMG_SIZE        = 224
MAX_SEQ_LEN     = 35
VOCAB_THRESHOLD = 3
TRAIN_RATIO     = 0.80
VAL_RATIO       = 0.10

# — Attention-specific —
ATTENTION_DIM   = 256 # size of attention hidden layer
ENC_FEAT_DIM    = 512 # CNN last stage channels
NUM_PIXELS      = 7 * 7 # spatial locations from CNN (7x7 feature map)

IMG_DIR = '/content/flickr8k/Images'
CAP_FILE = '/content/flickr8k/captions.txt'

print('Hyperparameters set.')
```

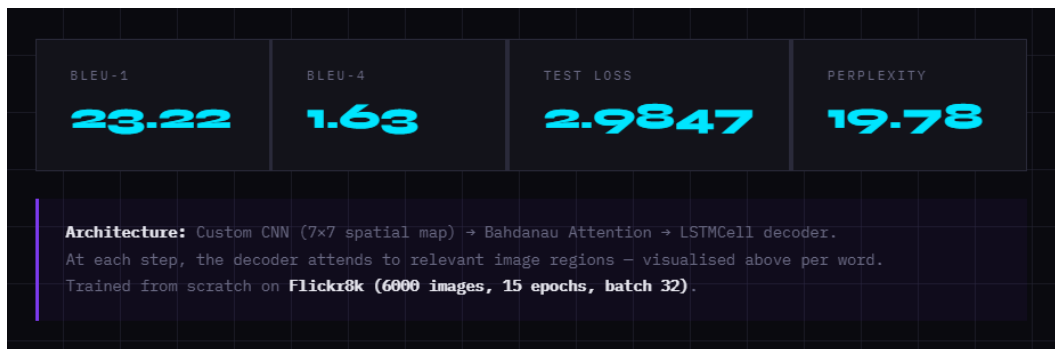
Evaluation Metrics

BLEU-1: 23.22

BLEU-4: 1.63

Test Loss: 2.9847

Perplexity: 19.78



```
Test Loss : 2.9847
Test PPL : 19.78
100%|██████████| 500/500 [00:07<00:00, 66.55it/s]
Avg BLEU-1 : 23.22
Avg BLEU-4 : 1.63
```

```
Metrics saved.
{'model': 'EncoderDecoder_Attention', 'test_loss': 2.9847, 't
```

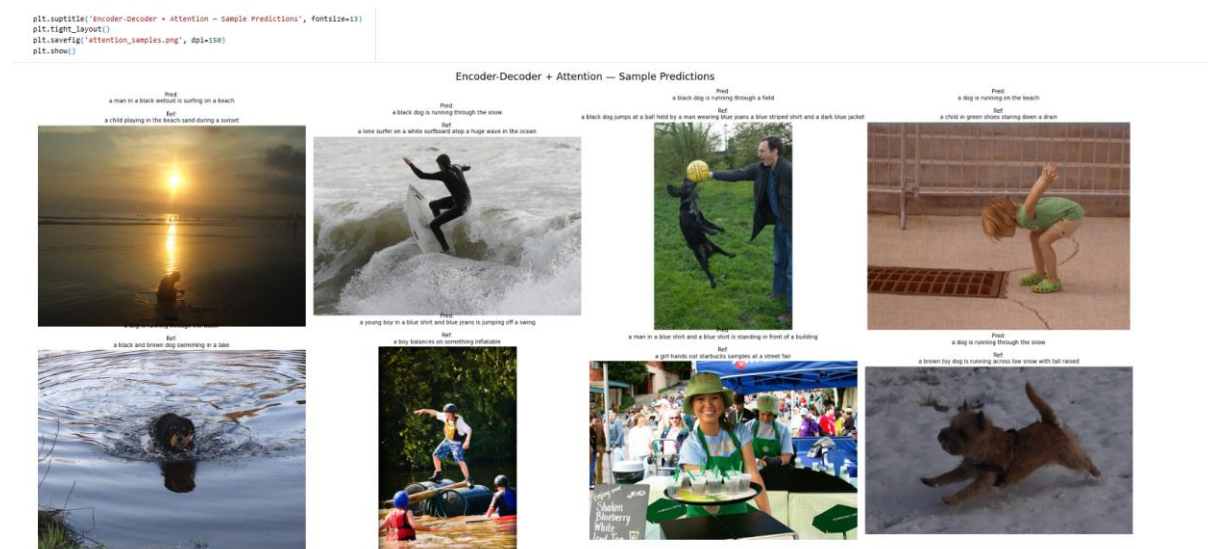
Displays the final evaluation metrics of the attention-based model, including test loss, perplexity, and BLEU scores.

```
print(f'=====')
print(f'{"Test Loss":<20} {"-":>20} {metrics["test_loss"]:>18}')
print(f'{"Test PPL":<20} {"-":>20} {metrics["test_ppl"]:>18}')
print(f'{"BLEU-1":<20} {"-":>20} {metrics["bleu1"]:>18}')
print(f'{"BLEU-4":<20} {"-":>20} {metrics["bleu4"]:>18}')
print('='*60)
print('Fill in No-Attention values from Colab 1 metrics_enc_dec.json')
```

```
=====
MODEL COMPARISON SUMMARY
=====
Metric                Enc-Dec (No Attn)    Enc-Dec + Attn
-----
(Run both notebooks and compare the printed values below)
Test Loss              -                2.9847
Test PPL               -                19.78
BLEU-1                 -                23.22
BLEU-4                 -                1.63
=====
Fill in No-Attention values from Colab 1 metrics_enc_dec.json
```

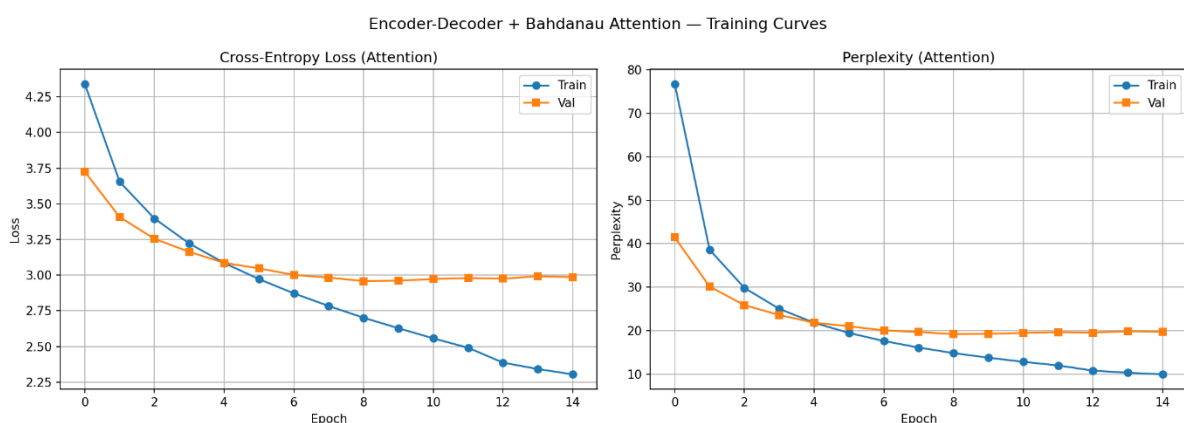
Compares performance of encoder-decoder models with and without attention, highlighting improved results using attention. The attention-based model shows improved BLEU scores and better contextual understanding compared to the encoder-decoder model without attention.

Lower perplexity and higher BLEU scores indicate more accurate, fluent, and contextually relevant caption generation.



The model shows reasonable performance in generating meaningful captions, though BLEU-4 is low due to dataset size and complexity.

Training Curves & Inference:

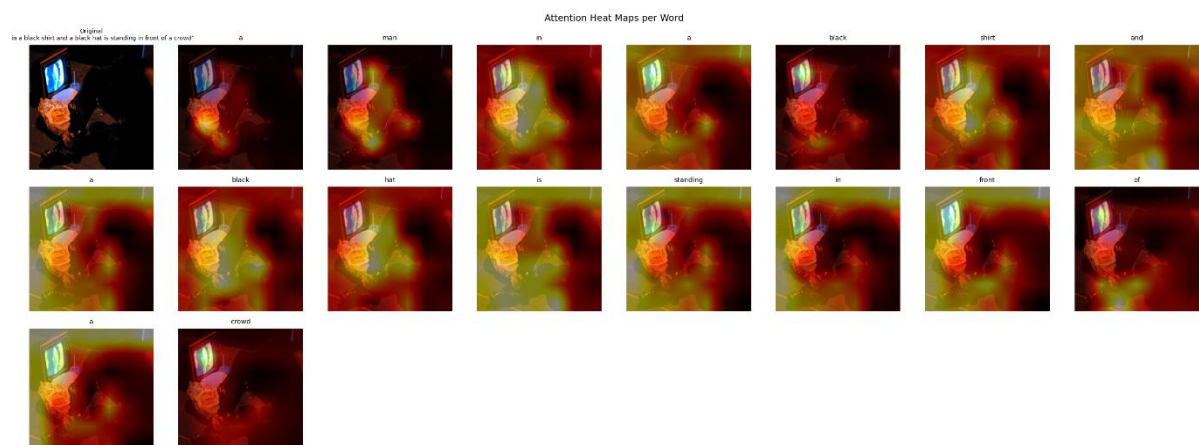
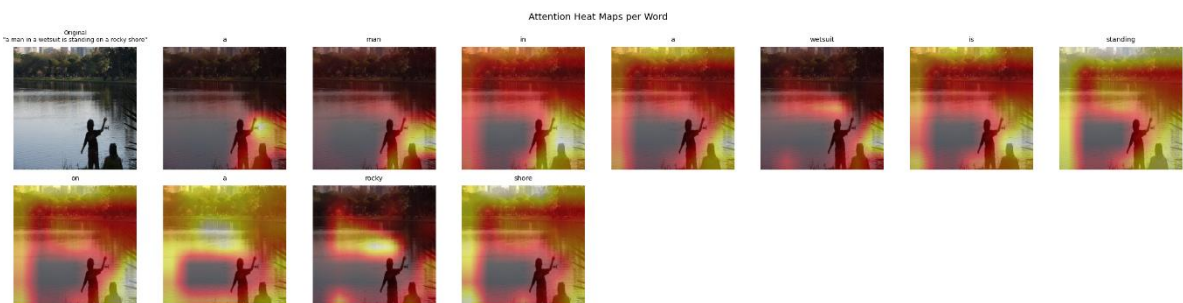
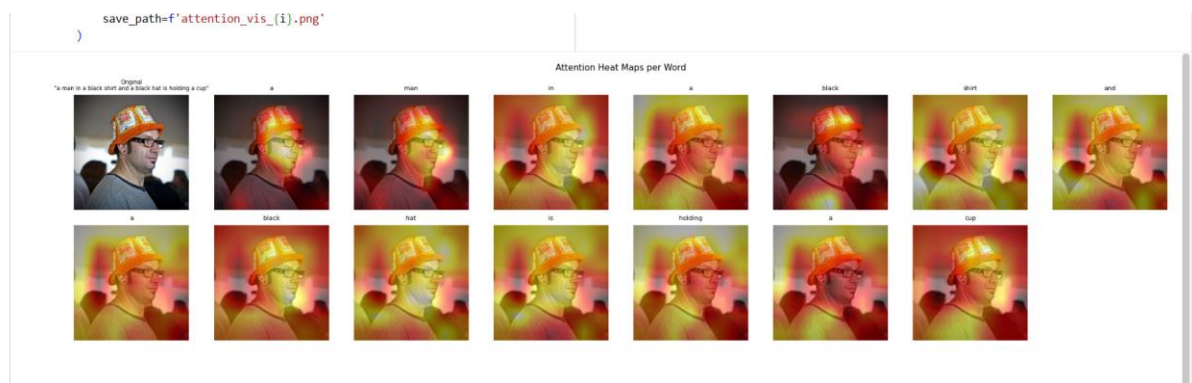


The graphs show the trend of cross-entropy loss and perplexity over training epochs for both training and validation data. Both metrics decrease steadily in the initial epochs, indicating effective learning. The validation curves stabilize after a few epochs, suggesting convergence and minimal overfitting.

This indicates that the model is learning meaningful image-caption relationships, and the attention mechanism improves caption relevance by focusing on important regions.

Attention Visualization

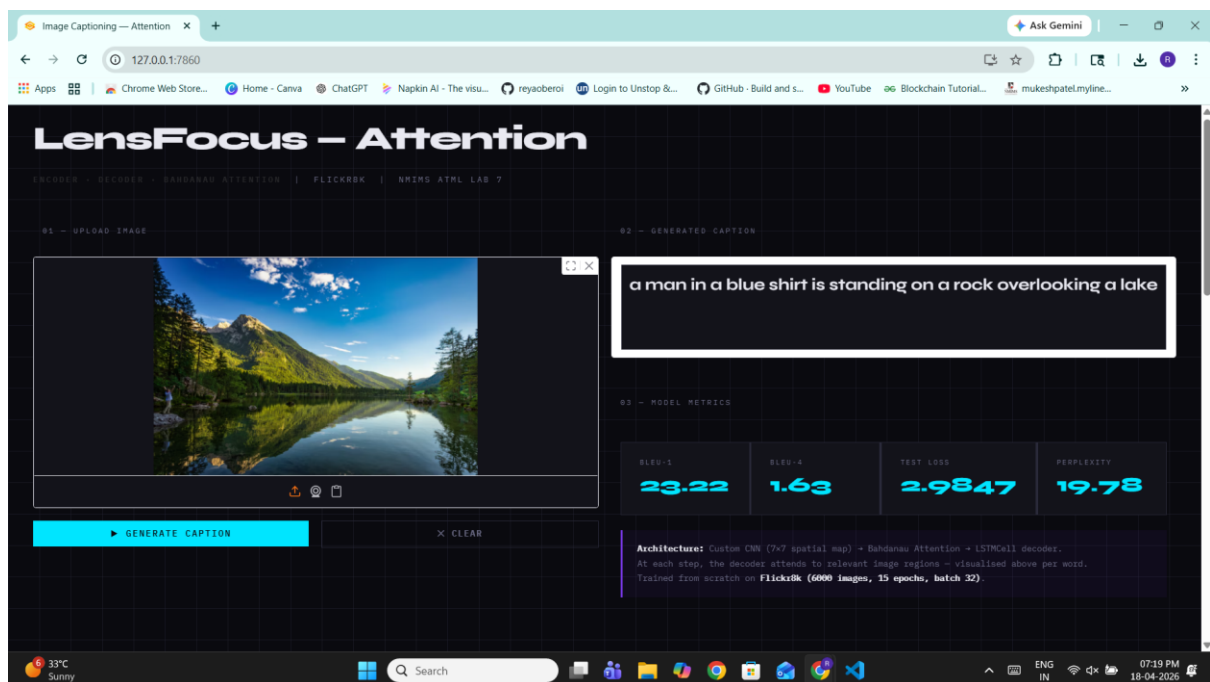
Attention maps highlight the regions of the image the model focuses on while generating each word. This improves interpretability and shows how visual and textual features are aligned during caption generation.



Integrating with Frontend (Deployment)

The project is implemented using modular code with separate files for model definition and deployment. The trained model is integrated with a Gradio interface for real-time caption generation.

- Used Gradio for interactive UI
- Image upload and caption generation functionality
- Displays generated caption instantly
- Shows attention heatmaps per word
- Metrics displayed for evaluation



Conclusion:

The attention-based model improves caption quality by focusing on relevant image regions during generation. While BLEU-4 is relatively low due to limited dataset size and caption variability, the model demonstrates correct semantic understanding. The project successfully integrates deep learning with an interactive UI.